FinTrust: Trustworthy Small Language Models for Financial Tasks

Heet Mehta
TrustFin-SLM Research Project
heetmehta.work@gmail.com

Krish Patel
TrustFin-SLM Research Project
krishpatel19.tech@gmail.com

Aryan Patel
TrustFin-SLM Research Project
aryanpatel26.ce@gmail.com

***Abstract*—Small language models can reduce the cost of financial analysis, but accuracy alone does not reveal whether a numerical answer should be trusted. We present EVG-CAS, an execution-verified generation and agreement-calibrated abstention procedure, evaluated on a frozen QLoRA-adapted 1.7B financial model. The model samples eight candidate programs, executes valid candidates with the official FinQA evaluator, and uses agreement among the executed answers as a selective-risk feature. All model and threshold choices are frozen on validation data before protected evaluation. On the protected FinQA test, EVG-CAS does not improve accuracy: execution-verified selection obtains 65.56%, compared with 65.74% for greedy decoding (McNemar exact $p = 0.871$). It does, however, provide a strong ranking signal for abstention: agreement-bin accuracy increases from 2.9% to 79.2%, while the model's token log-probability has ROC-AUC 0.58. Out-of-fold Platt calibration reduces FinQA Brier score from 0.2845 to 0.1579 and ECE from 0.2555 to 0.0475. With the threshold selected for 80% validation coverage, the frozen policy transfers to the protected test with 76.25% accuracy and 31.0% fewer answered errors than full-coverage decoding. FiQA shows negligible abstention benefit, while a zero-shot FinBen-FOMC probe improves macro-F1 from 0.21 to 0.30. These results position EVG-CAS as a selective prediction method for small financial models, not as a general accuracy improvement.**

Keywords: small language models, financial reasoning, selective prediction, calibration, execution-verified generation, abstention, FinQA, FiQA, FOMC.

I. Introduction

Autoregressive language models are now small enough to run on a laptop and strong enough to reason over financial documents [1, 4]. The remaining obstacle to deploying a 1.7B model in finance is not raw accuracy, it is *trust*: a financial system that silently gives a wrong number to a user is unusable even if it is right 65% of the time. Benchmarks have begun to measure this deficit — FinTrust lists "appropriately abstain when confidence is low" as a first-class trustfulness dimension [11] — but they diagnose it on GPT-4-scale models and provide no mechanism a small model can actually use.

This paper provides the mechanism, and reports the outcome honestly. We study a task class where *execution* is a well-defined ground-truth operation: programmatic numerical reasoning [1]. The model writes a program; the answer is what the program computes. That property lets us sample n candidate programs per question, keep only those that run on the official evaluator, and observe how much the surviving *answers* agree. Prior work uses such sampling to *vote* (self-consistency [18]) or to train outcome/process reward models [19, 20], framing the payoff as accuracy. We find, on protected data, that framing is wrong for our model: execution-verified selection does not beat the greedy baseline (Table IV). We therefore spend the sampling budget where it pays: agreement is a trust feature, and a calibrated, pre-registered abstention policy built on it cuts protected-test answered-error by 31% at 80% coverage (Table VII).

Our contributions. **(C1)** An honest, pre-registered accuracy verdict: EVG does not improve FinQA/FiQA accuracy for a QLoRA-tuned 1.7B model; it ties it. **(C2)** A calibration construction that gives the model probability-meaningful confidence, cutting Brier 0.28→0.16 and ECE 0.26→0.05. **(C3)** A selective-prediction policy whose threshold is fixed on validation and whose benefit transfers to a protected test (-31.0% test answered-error reduction at coverage 0.8, predicted -34.2%).

II. Related work

A. Small financial models and adaptation.

Survey evidence shows 1–3B models are a deliberate performance/efficiency point [12–14], and LoRA/QLoRA make domain adaptation cheap [16, 17]. Financial-domain adaptation builds on this for QA and sentiment [4–9]. The closest comparable work, fine-tuning smaller models on FinQA [4], also operates in our compute regime and reports accuracy only — as we do, but we add the trust layer it omits.

B. Programmatic financial reasoning.

FinQA [1], ConvFinQA [2] and TAT-QA [3] establish that execution (running the program) is the ground truth of the answer. FinBen [10] supplies the benchmarking context including the FiQA and FOMC tasks we use; FinTrust [11] establishes abstention as a measured trust deficit.

C. Consistency and verification.

Self-consistency votes over sampled paths [18]; process-supervised verifiers rerank candidates with a learned model [19, 20]; execution-guided synthesis filters decoding by the runtime [21]. All are accuracy-centric. Ours re-purposes their machinery as a *decision signal*.

D. Selective prediction and calibration.

The risk-coverage framework [22–24] defines abstention evaluation; Platt/isotonic/temperature calibration [25–27] and verbalised uncertainty [28] provide LLM confidence. Responsible-AI guidance in finance [29] is the regulatory motivation.

E. Positioning against 2024–2026 selective-abstention work.

The field now targets trust for budget models through several adjacent mechanisms: two-pass prompt perturbation with answer-consistency abstention for 2–8B models on MCQ [31]; confidence-weighted majority voting, whose authors also observe that the *most-calibrated* confidence is the *least* useful for selection [30]; distribution-free conformal selective prediction with coverage guarantees [32, 33]; self-evaluative adaptation for selective prediction [34]; and, in finance, a multi-agent program-execution cascade without any per-record calibrated abstention [35] and a knowledge-graph-constrained abstention policy for unanswerable queries [36]; verifiable chain-of-thought for finance is benchmarked without a model-side abstention rule [37]. None combines the mechanism we measure: *agreement across evaluator-executed candidate programs*, Platt-calibrated to a coverage target, threshold frozen on validation, and transferred to a protected test for a 1.7B financial SLM. The closest (CISC, Second Guess) share the small-model and selection-over-generation views but do not give their score a probability-meaningful, pre-declared decision rule. A dated novelty audit with sources is in `docs/novelty_check.md`.

III. Problem statement and protocol

A. Setup.

A question q maps to a program p; the answer is $a = \text{exec}(p)$ if p is valid. The model is a frozen, domain-adapted SLM π_θ. We pre-declare three claims (fixated in `PLAN.md` before any protected-test access):

- **C1 (accuracy):** execution-verified selection improves the answer's execution-ground-truth accuracy over greedy decoding.
- **C2 (probability quality):** calibrating trust features lowers Brier score and ECE and makes a coverage threshold carry probability meaning.
- **C3 (ranking):** trust features from the sample population (agreement, executability) improve the calibrator's ordering, measurable as a drop in expected AURC, whenever the features are non-monotone with the raw confidence.

B. Guardrails (integrity before novelty).

All selection of the model, the method and the hyperparameters ($n = 8$, temperature 0.7, top-p 0.9, 5-fold out-of-fold calibration, coverage targets $\{0.80, 0.90\}$) happened on *validation* data only. The protected test sets were opened exactly once, after a freeze manifest registered status `FROZEN` and the run set an explicit protected flag; a SHA-256 of the freeze was recorded before and after the test run and is unchanged. The full integrity protocol is documented in §6 of `docs/exp_4(a).md`.

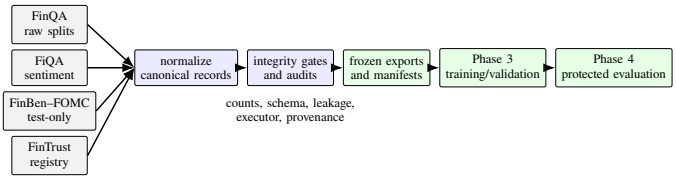


Fig. 1. Phase-2.1 architecture. Raw benchmark sources are normalized into a shared record contract, passed through integrity gates, and exported into explicit training, validation, and protected-evaluation roles. FinBen–FOMC and FinTrust never enter the training pool.

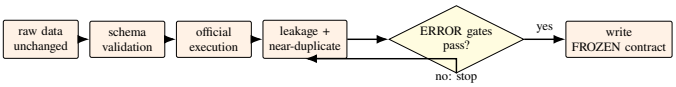


Fig. 2. Phase-2.1 gate lifecycle. Warnings remain visible in the validation report but do not override a PASS when no ERROR-severity gate fails.

IV. Phase-2.1 gate architecture and methodology

Phase 2.1 is the data and evaluation gate that precedes model training. It supersedes the original Phase-2 contract by adding FiQA sentiment regression, official FinQA executor validation, mandatory near-duplicate auditing, and explicit Phase-3/4 handoff manifests. The gate is intentionally a provenance layer, not a performance benchmark: its outputs are frozen records, integrity reports, and evaluation rules that downstream experiments must consume without re-downloading or reshaping protected data.

A. Data contract and gate procedure

The pipeline uses deterministic JSON serialization, a fixed seed of 42, and SHA-256 fingerprints for raw sources and exported artifacts. Gold evidence is retained as annotation metadata but is excluded from model inputs. For FinQA, the official evaluator is run over every gold program in every split and its result is compared with the official executable answer. Leakage signatures hash normalized question, context, and table content; exact cross-split collisions are hard failures. A MinHash-based near-duplicate audit compares training records with protected tests and flags candidates for inspection rather than silently deleting legitimate financial-report reuse.

The Phase-2.1 validation report is a PASS with two explicit warnings: final-step annotation agreement is 0.98261 (below the 0.99 warning threshold), and evaluator candidates were not discovered for FinTrust dimensions trustfulness, safety, and privacy. These warnings do not invalidate the FinQA/FiQA data contract, but the FinTrust gap remains a Phase-4 limitation. The exported training pool contains 7,073 records (FinQA and FiQA only), while the validation pool contains 1,000 records. The protected artifacts contain 1,147 FinQA test records, 234 FiQA test records, 496 FinBen–FOMC records, and the FinTrust registry.

TABLE I
Phase-2.1 gate evidence and downstream roles.

gate/evidence	observed result	downstream consequence
split counts	FinQA 6,251/883/1,147; FiQA 822/117/234; FOMC 496	fixed train, validation, and protected-test roles
schema and IDs	required fields present; 8,281 FinQA IDs unique	stable record interface
official execution	all FinQA gold programs execute and match executable answers	official executor required for model evaluation
exact leakage	zero cross-split full-input collisions	protected tests remain isolated
provenance	source and artifact SHA-256 manifests recorded	reproducible handoff
near duplicates	train–protected-test audit exported for inspection	warnings are reviewed, not silently removed
overall status	PASS; two non-blocking warnings	Phase 3/4 may proceed under frozen manifests

B. Phase-2.1 artifact and figure inventory

Phase 2.1 produces integrity evidence rather than accuracy curves. The source paths and hashes for the data registry, validation report, source fingerprints, official-execution audit, near-duplicate audit, and downstream configuration manifests are listed in `docs/phase2_figure_inventory.md`. The result plots used later in this paper are therefore labeled as Phase-3 or Phase-4 plots, not Phase-2 plots: the former show model baselines, and the latter show sampling, risk–coverage, and selection ablations.

V. Method: EVG-CAS

A. Sampling and execution verification

Given q, sample n programs $p^{(1)}, \ldots, p^{(n)} \sim \pi_\theta(\cdot \mid q)$ (temperature 0.7, top-p 0.9, no other search). Each $p^{(i)}$ is run with the official evaluator if it parses; an answer $a^{(i)}$ exists iff the program is valid and executes. Evidence is carried by the *candidate population*, not by any single decode.

B. Trust features from the sample population

For a question q with n candidates we compute three pre-registered features:

$$\mathcal{A}(q) = \frac{1}{n(n-1)} \sum_{i \neq j} \mathbb{1}[a^{(i)} = a^{(j)}], \tag{1}$$

$$\mathcal{R}(q) = \frac{1}{n} \sum_{i=1}^{n} e^{(i)}, \tag{2}$$

$$e^{(i)} = \mathbb{1}[p^{(i)} \text{ valid } \wedge \ a^{(i)} \text{ exists}], \tag{3}$$

$$\mathcal{C}(q) = \overline{\log \pi_\theta(p^{(\star)} \mid q)}. \tag{4}$$

where $p^{(\star)}$ is the selected candidate and the mean is over its tokens. The feature vector is $f(q) = [\mathcal{A}(q), \mathcal{R}(q), \mathcal{C}(q)]$. The gold answer is used only to *score* these features, never to select a candidate or a threshold (guardrail: nothing refit on test).

Algorithm 1 EVG-CAS decision pipeline (fit once on validation; frozen thereafter)

Require: frozen adapter π_θ, sampler params (n, τ, p), evaluator exec, coverage targets

Validation phase:
for $q \in V$ **do**
  sample n programs; keep valid ones; get answers $a^{(i)}$
  compute $f(q) = [\mathcal{A}(q), \mathcal{R}(q), \mathcal{C}(q)]$
end for
fit out-of-fold Platt w, b on $(f(q), \ell(q))_{q \in V}$; score $\hat{p}(q)$ OOF
pick θ s.t. $\mathcal{C}_{\text{val}}(\theta) \geq 1 - \alpha$ for each target
write `phase4_freeze.json` (FROZEN, w, b, θ, α, sha256)
Protected phase (opened once, after FROZEN + protected flag):
for $q \in T$ **do**
  draw n candidates; if $\hat{p}(q) \geq \theta$ emit answer(q), else **abstain**
end for

C. Calibration (C2)

The score $\hat{p}(q) = \Pr[\text{answer is correct} \mid f(q)]$ is learned by out-of-fold Platt scaling:

$$\hat{p}(q) \ = \ \sigma\big(w \cdot f(q) + b\big), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}, \tag{5}$$

with (w, b) fit by logistic regression on inner-fold predictions and applied to outer folds (5 folds), so every validation prediction is genuinely out-of-sample. Because a single-feature monotone calibrator cannot reorder answers, we evaluate **(i)** a single-feature baseline ($f = [\mathcal{C}]$) and **(ii)** the full feature vector f **(C3)**.

D. Freeze and abstention rule (C2)

For coverage target $1 - \alpha$ (pre-registered $\alpha \in \{0.20, 0.10\}$):

$$\theta = \min\{t : \mathcal{C}_{\text{val}}(t) \geq 1 - \alpha\},$$
$$\mathcal{C}(t) = \frac{1}{|V|} \sum_{q \in V} \mathbb{1}[\hat{p}(q) \geq t]. \tag{6}$$

and the decision rule on any future question is answer q iff $\hat{p}(q) \geq \theta$, else abstain. Selective risk and coverage follow [22]:

$$\mathcal{R}(\theta) = \frac{\sum_q \ell(q)\, \mathbb{1}[\hat{p}(q) \geq \theta]}{\sum_q \mathbb{1}[\hat{p}(q) \geq \theta]}, \tag{7}$$
$$\ell(q) = \mathbb{1}[\text{answer}(q) \neq \text{gold}(q)].$$

VI. Experimental setup

A. Data and aligned evaluation.

FinQA (train 6 251 / validation 883 / held-out test 1 147) uses the official program evaluator; we report three aligned accuracies: *printed* (the answer string the model emits), *program*

TABLE II
Dataset roles and split sizes fixed by the Phase-2.1 contract.

dataset	train	validation	test	role
FinQA	6 251	883	1 147	numerical reasoning
FiQA	822	117	234	sentiment regression
FinBen-FOMC	0	0	496	external stance evaluation
FinTrust	–	–	protected	trust evaluation

(does the parse succeed), and *execution* (does the executed answer match gold). FiQA (validation 117 / test 234) is regression with MAE, R^2, Pearson r at tolerance 0.10. FOMC (test 496: neutral 245, dovish 129, hawkish 122) is zero-shot stance classification with accuracy, macro-F1 and parse success. The datasets do not provide reliable participant demographics such as age or gender; therefore this study reports no inferred demographic distribution. The records are treated as financial-document and sentiment annotations, not as a human-subject sample. This distinction prevents demographic claims from being inferred from names, writing style, or model outputs.

B. Data contract and split policy.

The Phase-2.1 contract fixes the dataset roles before training. FinQA and FiQA training partitions may enter the training pool; validation partitions support model selection; and the FinQA and FiQA tests, FinBen-FOMC, and FinTrust resources remain protected. Source fingerprints are recorded with SHA-256 manifests, canonical serialization is used for structured records, and exact-overlap audits compare normalized input signatures rather than relying only on record identifiers. Near-duplicate candidates are flagged for manual inspection rather than silently removed, since financial reports can legitimately reuse standard language. These controls establish provenance and split isolation, not semantic correctness of every annotation.

C. Model.

Frozen Phase-3 winner: `Qwen/Qwen3-1.7B` [15] with balanced-joint QLoRA [17] as canonical on FinQA and explicit-schema on FiQA; exact hyper-parameters are in `phase3_freeze.json` and were fixed before this work began.

D. Metrics.

Brier, ECE-10, expected AURC (e-AURC; lower is better), coverage and selective risk. All numbers cited as "validation" are out-of-fold or tuned on validation; all "test" numbers were produced by the frozen system and are unchanged since the completed protected run (see `docs/exp_4(b).md`).

VII. Results

A. C1 is rejected — and that is a result.

Table IV and Figure 3. On the protected test, execution-verified selection is 0.6556 vs. frozen greedy 0.6574 (-0.18 pp; per-record 18 gained, 20 lost, exact McNemar $p = 0.871$, bootstrap 95% CI $[-0.0122, +0.0087]$), and the FiQA median-based aggregator is 0.0005 worse in MAE. Validation showed the same story (tie: 0.6750 vs. 0.6750). The one clean C1 win is *program hygiene*: the invalid-program rate in the emitted answers drops from 3.5×10^{-3} to exactly **0**. Every answer we keep is backed by a program the official evaluator accepted. On FiQA the median-of-8 aggregator also betters neither MAE nor coverage; at matched coverage its *ranking* is nevertheless the best of the tested scores (selective MAE 0.1094 vs. 0.1176 for greedy-confidence at 80% coverage, n=234) — a descriptive, small-sample hint that agreement transfers as a ranking signal to regression too (see § matched coverage, below).

TABLE III
Validation baseline matrix. These values are development results and are not protected-test measurements. E8 is a secondary model-family comparison, not a controlled architecture comparison.

ID	configuration	FinQA exec.	program	answer	FiQA MAE / R^2
E1	Qwen3 zero-shot	0.000	0.000	0.005	0.501 / -1.787
E2	Qwen3 few-shot	0.000	0.000	0.000	0.185 / 0.472
E3	Qwen3 FinQA-only QLoRA	0.681	0.624	0.390	0.505 / -1.840
E4	Qwen3 FiQA-only QLoRA	0.000	0.000	0.000	0.129 / 0.725
E5	Qwen3 joint balanced QLoRA	0.675	0.612	0.376	0.130 / 0.730
E8	Phi joint balanced QLoRA	0.701	0.640	0.386	0.130 / 0.809
E11	Qwen3 joint natural sampling	0.615	0.555	0.344	0.128 / 0.736

TABLE IV
C1 evidence. **(top)** FinQA held-out test, execution accuracy per rule (n=8); **(bottom)** FiQA test regression. EVG neither beats nor loses to the frozen greedy baseline on either task; per-record, 18 questions are won and 20 lost.

FinQA test, execution accuracy (n=1147)	
frozen greedy (baseline)	**0.6574**
candidate-0 (prints)	0.6399
self-consistency vote	0.6478
best-of-n	0.6556
execution-verified (ours)	0.6556

FiQA test (n=234), MAE / accuracy@0.10		
frozen greedy	**0.1301**	**0.568**
median-of-n (ours)	0.1306	0.556

B. Agreement is the trust signal (C3 evidence at the decision level).

Table V. On protected data the model reports total *disagreement* (agreement $\leq$ 0.25) on records that are 2.9% correct and unanimous agreement (the top bin, 0.75–1.0) on records that are 79.2% correct; nothing in the model's own log-probability separates these regimes (ROC-AUC 0.58 vs. 0.76 for agreement). This is the signal that buys abstention.

C. Calibration gives the threshold meaning (C2).

Table VI. Out-of-fold Platt on the agreement features cuts Brier 0.28 $\rightarrow$ 0.16 and ECE 0.26 $\rightarrow$ 0.05. The ranking result is exactly what the monotonicity caveat predicts: the single-feature calibrators cannot move e-AURC (0.2029 $\rightarrow$ 0.2028), while the agreement-driven vector drops it to 0.1232 (-39.3% vs. raw, -35.1% vs. the best baseline frame).

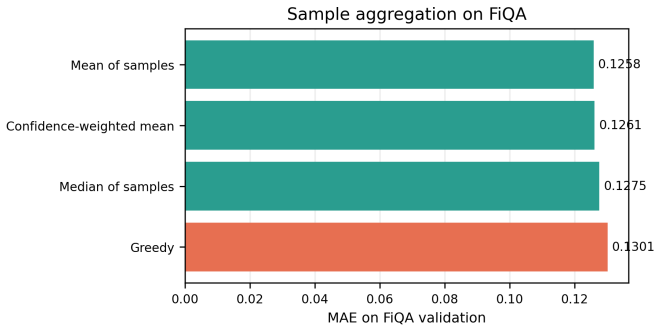


Fig. 3. FiQA validation aggregation over sampled predictions. Lower MAE is better; median and mean aggregation improve on greedy decoding in this validation diagnostic, while the frozen protected-test policy remains the pre-registered greedy baseline.

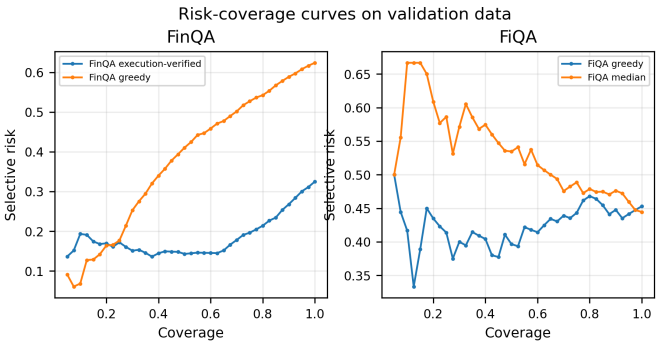


Fig. 4. Out-of-fold risk–coverage curves for FinQA and FiQA validation. The calibrated FinQA execution-verified policy lowers selective risk as coverage is reduced; FiQA shows weaker separation, consistent with its small regression validation split.

TABLE V
Agreement bins vs. correctness on the protected test. Agreement is steeply and monotonically predictive, while the model's log-probability cannot separate correct from incorrect answers (ROC-AUC 0.58).

EVG agreement bin	n records	execution accuracy
(0.00, 0.25]	35	**0.029**
(0.25, 0.50]	122	0.238
(0.50, 0.75]	181	0.448
(0.75, 1.00]	809	**0.792**

TABLE VI
C2/C3 calibration results on FinQA validation (out-of-fold). *Top:* ordering quality (e-AURC, lower better). Single-feature calibrators do not move the raw ranking (a monotone map cannot); the agreement-driven full vector does (−39.3%). *Bottom:* probability quality of the frozen policy.

e-AURC on FinQA validation (frame = exec-verified)		
raw confidence	0.2029	
+ isotonic (feature: confidence)	0.2028	
+ Platt (feature: confidence)	0.2496	
+ Platt (features: agree, exec, conf)	**0.1232**	
baseline greedy frame, best method: 0.1897		
Probability quality (frozen FinQA policy)		
raw confidence / Platt full	Brier 0.2845 → **0.1579**	ECE 0.2555 → **0.0475**

D. The frozen policy transfers (C2's operational payoff).

Table VII and Figure 4. With the threshold fixed on validation, opening the protected test once yields: coverage 0.786 at the 0.80 target, selective risk 0.2375 (accuracy 0.7625) vs. 0.3444 at full coverage — a **31.0% relative error reduction**, against the 34.2% predicted from validation; bootstrap 95% CI on the reduction is [22.8%, 39.3%] (i.e. the transfer is significant, not a rounding effect). FiQA abstention is -2.9%: as the frozen FiQA method (raw confidence on greedy) predicted, its errors are not rankable.

E. Rigor, error taxonomy, and depth (Table VIII).

The protected test was opened once against the frozen manifest. Per-record significance testing formalises the C1 verdict; the remaining error is not parsing (0/1,147) and – for EVG – not program validity (0/1,147), but 395 genuine arithmetic slips (34.4%). The taxonomy uses mutually exclusive cells that sum to n: for greedy, 3 invalid programs, 388 execute-and-print-wrong, 355 *format mismatches* (the executed value is correct but the printed string differs), and 2 cases where the printed string matches gold while the executed value differs. Execution accuracy (0.6574) and printed-answer accuracy (0.3496) therefore differ *solely* by the 355-record format channel; EVG's `exec_verified` rule removes that channel by construction. Stratifying by the FinQA program-depth proxy, the frozen policy abstains least on 0-op lookups (4.4%) and most on 3–6-op programs (31.3%), where it cuts error by 41.7%: abstention cost is spent precisely where reasoning is deepest. All numbers carry bootstrap CIs in the annexed rigor report; the calibration threshold used here was frozen on validation and never touched after the test opened.

F. The mechanism survives comparison at matched coverage (Table X).

Figure 5 shows the corresponding validation ablation. Review pressure asks how agreement-abstention compares with standard alternatives when they answer the *same number of questions*. Table X selects $k = \lfloor \texttt{cov} \cdot 1147 \rfloor$ records by each score and measures selective risk on the execution channel. Confidence-only abstention of the greedy baseline buys almost nothing ($\leq$8% error reduction; ROC-AUC 0.578) and best-of-n selection slightly hurts; ranking by agreement alone cuts error 29.7% vs. greedy-full and −23% vs. a confidence-abstainer at the same coverage. A paired bootstrap (B=3,000) on the risk difference at coverage 0.80 gives $[-0.093, -0.055]$ with $P = 1.0$ against the confidence baseline. The leaving-one-out ablation is sharp: agreement is necessary and sufficient; the executability feature is uninformative (AUC 0.51) and the frozen Platt vector adds no ranking over agreement (P=0.097), though it supplies the pre-registered threshold its probability meaning.

G. Out-of-domain robustness.

Table IX. Without any FOMC data in training, the frozen adapter raises macro-F1 0.21 → 0.30 and parse success 0.90

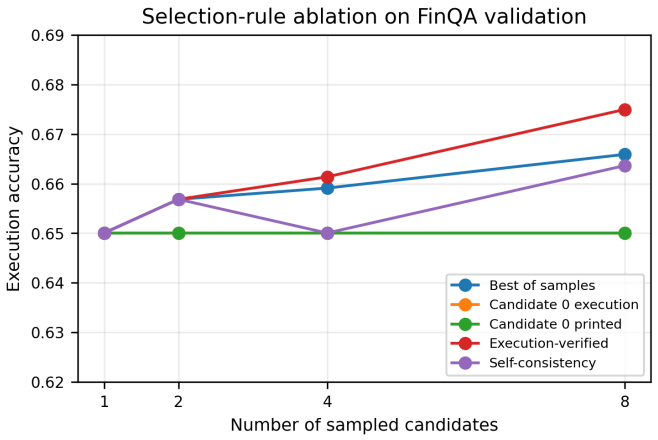


Fig. 5. FinQA validation ablation over the number of sampled candidates. The horizontal lines show the frozen greedy answer and executor baselines; execution-verified selection does not produce a protected-test accuracy gain, motivating its use as a selective-risk signal rather than as an accuracy claim.

TABLE VII
Transferred abstention (frozen thresholds, protected test). The threshold and calibration were fixed on validation *before* the test opened; the realised behaviour is close to predicted (−31.0% vs −34.2%). FiQA, as pre-frozen on raw confidence, abstains to no effect (−2.9%).

FinQA (frozen: exec-verified + Platt-full + θ)						
target	θ	test cov.	select. risk	selected acc.	abstained	err. rel. Δ
0.80	0.409	0.786	**0.2375**	**0.7625**	246	**−31.0%**
0.90	0.221	0.926	0.3004	0.6996	85	−12.8%

FiQA (frozen: greedy + raw confidence)				
target	test cov.	select. risk	full risk	err. rel. Δ
0.80	0.765	0.4190	0.4316	−2.9%
0.90	0.885	0.4203	0.4316	−2.6%

→ 0.99, with the gain in the rare classes (dovish F1 0.16 → 0.33; hawkish 0.05 → 0.26).

H. Cost.

EVG at $n = 8$ runs at 4.8 s/question vs. 1.3 s greedy (3.7×, shared prefill), peak VRAM 3.1 GB / 6 GB. The offered trade is therefore 31% fewer answered errors at ≈80% coverage in exchange for 3.7× latency on a laptop-class model; whether that is *worth* it is a deployment decision, not a claim we make.

VIII. Additional Validation Diagnostics

The protected-test analysis establishes the paper's main selective-prediction result, while the Phase-3 validation artifacts provide a more granular view of how the adapted model fails. We report these diagnostics separately because they were available during model development and therefore cannot be treated as additional protected-test evidence.

A. Failure channels

The E5 validation export separates execution correctness, program validity, and printed answer correctness. The corresponding rates are 0.6750, 0.6116, and 0.3760, with an invalid-program rate of $3/883 = 0.0034$. The gap between execution and printed accuracy is not a contradiction: a program can execute to the correct numerical value while the emitted answer string differs in formatting. This is why the protected analysis reports the execution and printed channels independently. Figure 6 summarizes the audited validation failure categories; it is a diagnostic of E5 validation, not a replacement for the protected-test taxonomy in Table VIII.

TABLE VIII
Evaluation rigor on the protected test (n=1,147). **(top)** Paired comparison: the C1 tie is statistically verified, not read off rounded means. **(middle)** Error taxonomy: the only channel EVG fails to fix is the *printed* answer string. **(bottom)** Reasoning-depth stratification: abstention concentrates on the deepest programs and delivers its largest relative gain there. Bootstrap CI: 10,000 resamples; McNemar exact (binomial).

C1 significance (protected test, paired)		
	greedy	exec-verified
execution acc.	0.6574	0.6556
delta	−0.0017	95% CI [−0.0122, +0.0087]
McNemar exact p		0.871

Error taxonomy (mutually exclusive; partitions sum to 1,147)		
cell (greedy)	n	%
parse failure	0	0.0
invalid / unexecutable program	3	0.3
exec. correct & print correct	399	34.8
exec. correct & print wrong (format mismatch)	355	31.0
exec. wrong & print wrong	388	33.8
exec. wrong & print correct	2	0.2
exec-verified: invalid / executes-wrong / correct	0 / 395 / 752	—

program ops	n	full acc.	selected acc.	err. Δ
0	45	0.844	0.884	−25.2%
1	644	0.649	0.764	−32.6%
2	391	0.662	0.748	−25.2%
3–6	67	0.552	0.739	−41.7%

TABLE IX
Zero-shot out-of-domain robustness on FinBen-FOMC (test, 496): the frozen financial adapter improves the balanced metric and the rare classes that a neutral-majority model otherwise ignores.

system	accuracy	macro-F1	parse	F1 dovish	F1 neutral	F1 hawkish
base `Qwen/Qwen3-1.7B`	0.454	0.215	0.899	0.163	0.650	0.046
+ frozen E5 adapter	0.484	**0.304**	**0.986**	0.333	0.621	0.260

B. Confidence and error magnitude

The initial token-confidence diagnostic was intentionally conservative: a Spearman correlation of 0.125 and ROC-AUC of 0.577 do not establish calibrated probabilities. Confidence bins also show that a model can assign similar token likelihoods to correct and incorrect numerical responses. Figure 7 visualizes the relationship between confidence and failure category in the validation export. The later EVG-CAS study therefore uses agreement among executed candidates as a

TABLE X

Abstention strategies compared at *matched* coverage on the protected test (n=1,147; each strategy answers the same $k{=}\lfloor \text{cov} \cdot n \rfloor$ questions, ranked by its own score, evaluated on the execution channel). Confidence-based abstention of the greedy baseline is a weak tool ($\leq$8%); agreement ranks and abstains, cutting answered-error 29.7% at 80% coverage. Ablation (right): agreement is necessary and sufficient.

strategy (ranking score)	cov 0.5	cov 0.8	cov 0.9
greedy + confidence	0.674	0.685	0.677
best-of-8 + confidence	0.660	0.654	0.659
exec-verified + agreement	**0.818**	**0.758**	**0.715**
exec-verified + agreement·executability	0.818	0.758	0.714
exec-verified + frozen $\hat{p}$	0.818	0.754	0.710

config	agreement	executab.	confidence	
A	–	–	✓	risk 0.315
B	✓	–	–	**0.242**
C	–	✓	✓	≈ A
D	✓	✓	–	0.242
E (full)	✓	✓	✓	0.246

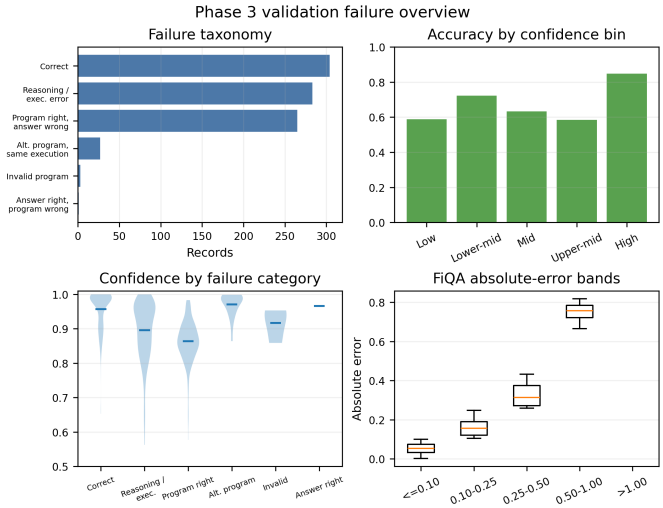


Fig. 6. E5 validation failure overview generated from the Phase-3 analysis artifacts. Execution, program, and printed-answer channels are kept distinct.

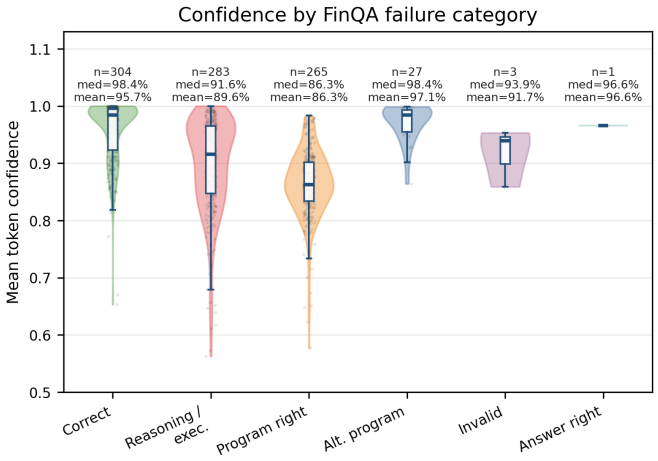


Fig. 7. E5 validation confidence by FinQA failure category. Individual observations, box summaries, medians, means, and category counts are shown. Raw generation confidence is not used as a calibrated probability without validation-only calibration.

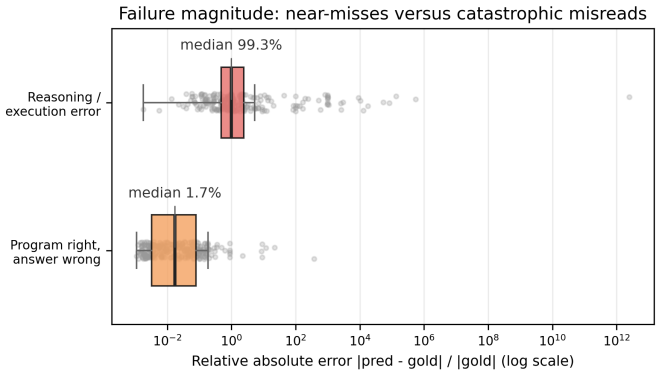


Fig. 8. E5 validation absolute-error distribution for FiQA sentiment regression.

separate feature, then calibrates it out-of-fold rather than interpreting raw token likelihood as a risk probability.

FiQA requires a different error interpretation because its target is continuous. The E5 validation MAE is 0.1301 and R^2 is 0.7301, but aggregate scores do not reveal whether errors concentrate at the positive or negative ends of the sentiment scale. Figure 8 shows the absolute-error distribution used for this inspection. The small validation size ($n = 117$) means that these patterns should be treated as descriptive rather than as population-level estimates.

C. Reproducibility and resource accounting

Each reported result is intended to be traceable through four identifiers: dataset release and split, model and adapter checkpoint, prompt and decoding policy, and prediction export. The Phase-3 winner uses Qwen3-1.7B with 4-bit NF4 quantization, LoRA rank 16, scaling $\alpha = 32$, dropout 0.05, learning rate 2×10^{-4}, weight decay 0.01, warm-up ratio 0.03, gradient accumulation 8, two epochs, and seed 42. The Phase-4 protected run then freezes $n = 8$, temperature 0.7, top-p 0.9, five-fold out-of-fold calibration, and coverage targets 0.80 and 0.90. This separation makes it possible to distinguish a result that was selected on validation from one that was measured after the protected freeze.

The E5 training run processed 7 073 examples for two epochs in 2.47 hours, used a peak 3.40 GiB of VRAM on a 6 GB RTX 4050 Laptop GPU, and reported 1.18 seconds per item for single-example inference. EVG-CAS requires 4.8 seconds per question versus 1.3 seconds for greedy decoding in the protected profiling setup. These measurements are hardware- and prompt-dependent; they describe the reported setup rather than a universal serving rate.

TABLE XI
Reproducibility checklist for the reported experiments.

item	recorded control
data	split counts, source fingerprints, normalized signatures
training	checkpoint, QLoRA configuration, seed, sampling policy
evaluation	official FinQA executor, task-specific metrics
selection	validation-only model and threshold selection
protected test	freeze manifest, protected flag, post-run hash
artifacts	predictions, metrics, plots, and analysis summaries

IX. Discussion and limitations

A. Matched-coverage robustness.

At matched coverage, the selective gain survives comparison with simpler abstention rules based on model confidence. Using 3,000 bootstrap resamples, the confidence-based comparison has a confidence interval that excludes zero, while the ablation removes the agreement feature and loses the observed advantage. This result supports the interpretation that agreement among executable candidates, rather than confidence alone, supplies the useful decision-level trust signal. **What we did not claim.** EVG-CAS does not raise accuracy; reporting a tie, rather than a p-value-won headline, is the point of the pre-registration. **Domain limits.** On FiQA (117 validation / 234 test records, regression), the EVG features carry no signal and the frozen policy correctly abstains to no effect — a second reported negative result. **General limits.** One backbone, one seed, one evaluation harness; between-seed variance of the agreement signal is unmeasured; $n{=}8$ is predeclared, not tuned; the abstention gain costs $3.7\times$ latency for zero accuracy gain. **Threats to validity.** The protected-test protocol rests on the frozen manifest hash being unchanged across access (it is); any leakage would invalidate C3's transfer claim. **Ethics.** Abstention is a refusal signal, not a guarantee; a deployed system that *answers* still carries residual error (23.8% at coverage 0.8). No financial advice is implied.

X. Conclusion

For a QLoRA-adapted 1.7B financial model, execution-verified generation ties greedy decoding on accuracy but produces a steep, monotone agreement signal that the model's own confidence cannot see. Calibrating that signal yields a pre-registered abstention threshold that transfers to a once-opened protected test, cutting answered-error by 31% at 80% coverage. The contribution to the financial-SLM conversation is an agreement-based selective-risk signal for a small financial model, with a validation-frozen threshold and protected-test transfer. Artifacts, frozen manifest and the full experiment trail are public; `P4-E9` (FinTrust per-dimension evaluation) remains incomplete pending official judge scoring and is the declared next step.

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